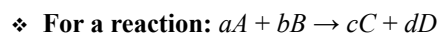


Chemical Kinetics

SHORT NOTES

❖ **Rate of reaction (ROR)** = $\frac{\text{Rate of disappearance of reactant (appearance of products)}}{\text{Stoichiometric coefficient of reactant (products)}}$



❖ **Instantaneous rate:** $-\frac{1}{a}\left(\frac{d[A]}{dt}\right) = -\frac{1}{b}\left(\frac{d[B]}{dt}\right) = \frac{1}{c}\left(\frac{d[C]}{dt}\right) = \frac{1}{d}\left(\frac{d[D]}{dt}\right)$

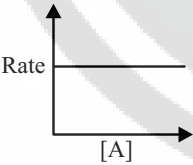
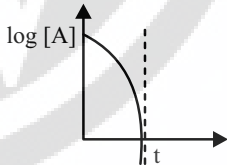
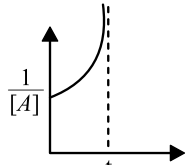
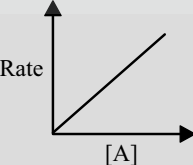
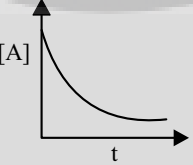
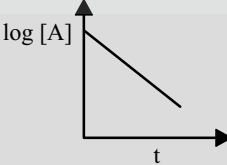
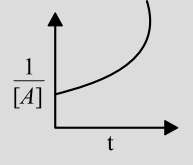
Relationship between rate of reaction and rate of disappearance of reactant (rate of appearance of product).

❖ **Average rate:** $-\frac{1}{a}\left(\frac{\Delta[A]}{\Delta t}\right) = -\frac{1}{b}\left(\frac{\Delta[B]}{\Delta t}\right) = \frac{1}{c}\left(\frac{\Delta[C]}{\Delta t}\right) = \frac{1}{d}\left(\frac{\Delta[D]}{\Delta t}\right)$

Table: Important kinetic expression for reaction of type A → B:

Order	Zero	1st	nth
Differential rate law	Rate = k	Rate = $k[A]$	Rate = $k[A]^n$
Integrated rate law	$[A_0] - [A] = kt$	$kt = \ln \frac{[A]_0}{[A]}$	$kt = \frac{1}{(n-1)} \left[\frac{1}{[A]^{n-1}} - \frac{1}{[A]_0^{n-1}} \right]$
Half life ($t_{1/2}$)	$t_{1/2} = \frac{[A]_0}{2k}$	$t_{1/2} = \frac{\ln 2}{k}$	$t_{1/2} = \frac{1}{k(n-1)} \left[\frac{2^{n-1} - 1}{[A]_0^{n-1}} \right]$
$t_{3/4}$	$t_{3/4} = 1.5 t_{1/2}$	$t_{3/4} = 2 t_{1/2}$	$t_{3/4} = (2^{n-1} + 1) t_{1/2}$

Table: Graphs of Zero and First Order Reactions

Order	Rate vs $[A]$	$[A]$ vs t	$\log [A]$ vs t	$\frac{1}{[A]}$ vs t
Zero order				
First order				

Important Characteristics of First Order Reaction

- ❖ $t_{1/2}$ is independent of initial concentration.
- ❖ In equal time interval, reactions finishes by equal fraction.

	$t = 0$	$t = t$	$t = 2t$	$t = 3t, \dots$
Reactant conc.	a_0	$a_0 x$	$a_0 x^2$	$a_0 x^3, \dots$

Zero Order

- ❖ $t_{1/2}$ of zero order is directly proportional to initial concentration.
- ❖ In equal time interval, reaction finishes by equal amount.

$t = 0$	$t = t$	$t = 2t$	$t = 3t, \dots$
C_0	$C_0 - x$	$C_0 - 2x$	$C_0 - 3x, \dots$

- ❖ Graph of $[A]$ vs t is straight line.

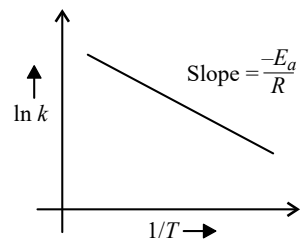
A zero order reaction finishes in $t = \frac{[A]_0}{k}$

Temperature Dependence

- ❖ Temperature coefficient = $\frac{k_{T+10}}{k_T}$
- ❖ Arrhenius equation: $k = A.e^{-E_a/RT}$
- ❖ A and E_a are constants i.e. do not vary with temperature

$$\ln k = \ln A - \frac{E_a}{RT}$$

Graph: Graphical determination of E_a .



By default, $T = 298K$

Variation of rate constant with temperature

$$\Rightarrow \ln \frac{k_2}{k_1} = \frac{E_a}{R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

